

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
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    "header": {},
    "path": {},
    "query": {
      "area": {
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        "required": false,
        "schema": {
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          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
            },
            "date_to": {
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            "required": false,
            "schema": {
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              "type": "string"
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            "required": false,
            "schema": {
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            "schema": {
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              "type": "boolean"
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            "required": false,
            "schema": {
              "type": "string"
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            "source": {
            "nullable": true,
            "required": true,
            "schema": {
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              "type": "string"
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                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
                      "occurred_at": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
                      "parameter_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
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                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                          "type": "integer"
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                        "nullable": false,
                        "required": true,
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                          "type": "string"
                        },
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                        "required": false,
                        "schema": {
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                          "type": "integer"
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                        "required": true,
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                          "type": "string"
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                        "schema": {
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                        "schema": {
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                          "type": "integer"
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                        },
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                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                          "min_length": 1,
                          "type": "string"
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                              "required": false,
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                                            },
                                            {
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                              }
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                          }
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                    }
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                }
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        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      }
    },
    "responses": {
      "200": {
        "schema": {
          "items": {
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            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                },
              "type": "object",
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                      "required": false,
                      "schema": {
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                          "fields": {
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                                      "type": "integer"
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                                    },
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                                  "type": "array",
                                  "shape": "object"
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                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

```
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```

04 POST /documents/search

RULES []

```
{request: {body: {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": {"type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "object"}, "array": true, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{request: {body: null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {"include_production_context": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}}, "production_context": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"returned_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "truncated": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{request: {body: null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id"
```

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  "nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}},
"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nu
  llable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_
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    : {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"
      }}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED",
        "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object", "type": "array"}}, "fau
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          ype": "string"}}, "fault_color": {"nullable": false, "required": true, "schema": {"type": "null"}}, "fau
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              lse, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "
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                        "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": tr
                          ue, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required":
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                              rue, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema":
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                                      y"}}, "type": "object", "type": "array", "shape": "array", "422": {"schema": {"additional_properties"
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                                          erties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "unio
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                                                {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}},
                                                07 POST /agent/ask
```

```
RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"}]
{"request": {"body": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": true, "requ
  ired": false, "schema": {"min_length": 1, "type": "string"}}, "question": {"nullable": false, "required":
    true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object", "header":
      {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields":
        {"answer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confide
          nce": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}},
            "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"
              chunk_id": {"nullable": false, "schema": {"min_length": 1, "type": "string"}}, "doc_id"
                : {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"
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                            le": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false
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                                            : 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1,
                                              "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type":
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```

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ma": {"min_length": 1, "type": "string"}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}}, "type": "array"}, "limit": {"nullable": false, "required": true, "schema": {"type": "string"}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object", "shape": "object", "404": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}, "503": {"schema": {"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}}, "details": {"nullable": false, "required": false, "schema": {"additional_properties": true, "fields": {"message": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "shape": "object"}}}
```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item s": {
            "additional_properties": false,
            "fields": {
              "action_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "EQP_HOLD",
                    "MONITORING",
                    "WARNING"
                  ],
                  "type": "string"
                },
              "action_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "agent_run_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approval_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approved_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "approved_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "chamber": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "created_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "decision_comment": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "equipment": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "equipment_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "lot": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "lot_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "predicted_fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "reason": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "status": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "APPROVED",
                    "PENDING",
                    "REJECTED"
                  ],
                  "type": "string"
                },
              "type": "object",
              "type": "array",
              "shape": "array"
            }
          }
        }
      }
    }
  }
}
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "decided_by": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 40,
            "min_length": 1,
            "type": "string"
          }
        },
        "decision": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["APPROVED", "REJECTED"],
            "type": "string"
          }
        },
        "decision_comment": {
          "nullable": true,
          "required": false,
          "schema": {
            "max_length": 1000,
            "type": "string"
          }
        },
        "object": {
          "header": {},
          "path": {
            "approval_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "query": {},
            "responses": {
              "200": {
                "schema": {
                  "additional_properties": false,
                  "fields": {
                    "action_code": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                        "type": "string"
                      }
                    },
                    "action_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "agent_run_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approval_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approved_at": {
                      "nullable": true,
                      "required": true,
                      "schema": {
                        "format": "date-time",
                        "type": "string"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ring"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}
```

```
{}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "recent_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, 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```

13 GET /alarms/{source}/{alarm_id}

RULES []

```
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```

```
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```

14 GET /alarms/paged

RULES []

```
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```

15 GET /traces/catalog

RULES []

```
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```

```
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```

16 POST /traces/search

RULES []

```
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```

17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
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```



```
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```

19 POST /agent/runs

RULES []

```
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```

20 GET /agent/runs/{run_id}

RULES []

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22 POST /agent/runs/{run_id}/retry

RULES []

deferred inventory

23 GET /agent/runs/paged

RULES []

deferred inventory

24 GET /approvals/paged

RULES []

deferred inventory

25 GET /actions

RULES []

{ "request": { "body": null, "header": {}, "path": {}, "query": { "action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"} } } }, "responses": { "200": { "schema": { "items": { "additional_properties": {"fields": { "action_code": {"nullable": false

[illegible]

```
{}}, "type": "object"}, "shape": "object"}}
27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
deferred inventory
28 POST /internal/actions/{action_id}/delivery
RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
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29 POST /analytics/query
RULES []
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RULES ¶

31 POST /analytics/validate

32 GET /analytics/history

33 GET /analytics/evaluations

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RULES []
{
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      },
      "page": {
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        "required": false,
        "schema": {
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      },
      "size": {
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        "required": false,
        "schema": {
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          "type": "integer"
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    },
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    }
  }
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34 GET /audit-logs/paged

RULES []

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[illegible]

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AgentEvaluationResponse

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AgentPredictionDetailItem

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AgentRunAccepted

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AgentRunAcceptedResponse

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AgentRunActionItem

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AgentRunApprovalItem

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AgentRunAskEvidence

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AgentRunCreateRequest

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AgentRunDetailResponse

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AutoToolCallItem

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ChamberGraphResponse

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ChamberRelationResponse

```
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ChartType

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ChatToolCallItem

```
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CrossCheck

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ring"}}, "type": "object"}
DeliveryCallbackRequest
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DeliveryChannel
{"enum": ["EMAIL", "MES MOCK"], "type": "string"}
DeliveryResult
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DeliveryStatus
{"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}
DiagnosisBlock
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DocumentChunkItem
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DocumentDetailResponse
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DocumentHit

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{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"pattern": "^.+cs2:[0-9]{4}$", "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1, "minimum": -1, "type": "number"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
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DocumentSearchRequest

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DocumentType

```
{"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}
```

ErrorCode

```
{"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}
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ErrorResponse

```
{"additional_properties": false, "fields": {"code": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "details": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "message": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
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EvaluationListResponse

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EvidenceAssessmentBlock

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EvidenceItem

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FaultHypothesis

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GraphAskEvidence

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GraphContext

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GraphEvidence

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GraphNode

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GraphQueryRequest

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GraphQueryResponse

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GraphRelationship

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GroupeMetricResult

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HTTPValidationError

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HealthResponse

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ImpactScopeBlock

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ImpactScopeItem

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MetricPlan

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MetrologyAskEvidence

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MetrologyEvidence

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NIQueryHistoryResponse

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NIQueryLogItem
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ParameterItem
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PostActionObservationBlock
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PublicAgentRunItem
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PublicApprovalDecision

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HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
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